



Tathastu Workshops
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DSA Workshop

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Overview of Session



- Role of DSA in Jobs/Internships
- How to choose a language (5 mins).
- How to study that language (5 mins).
- Learning DS & A (20 mins).
- CP Roadmap (10 mins).
- Some recommended resources (5 mins).
- Sample Problems (25 mins).

Role of DSA in Jobs/Internships



- < 4 LPA (IT/Service Based Companies) : Basic knowledge is required for MCQ and Interviews. Upto basics of Tree.
- 5 - 12 LPA (Product Based Companies) : Good command on DSA, required upto Tree, Graphs and even DP.
- 12 LPA + (Tier 1 Companies) : Excellent knowledge of DSA, good profile on CP platforms and knowledge of all advance DSA topics is required.

How to choose a language?



- You only have 3 options : C++, JAVA and Python.
- C++ : Useful in written of companies, not much projects to be made, best suited for CP.
- JAVA : (recommended) Helpful in job interviews, web and android projects, plenty of online support.
- Python : If you wish to make career in Data Science, you can make many projects in it.

Learning a Language



- Choosing the language.
- Learn just basic syntax, loops, data types, functions, recursion, etc.
- Practice few programs like prime number, palindrome, string operations, printing patterns, etc.
- You can use Hackerrank for practice.
- Books for different languages.

Linear Data Structure



- **Array** : Practice simple problems in it, at least 20-25 like reversal, sort, rearrange, negative numbers, etc.
- **Stack** : Learn implementation, practice 10-15 questions like operations, prefix to postfix, expression evaluation, etc.
- **Linked List** : Learn implementation, it's types, practice 25-30 questions like reversal, detect loop, middle, etc.
- **Queue** : Learn implementation, it's types, practice 10-15 questions like LRU, page replacement, sliding window, etc.

Non Linear Data Structure



- Tree : Binary Tree, Binary Search Tree, implementation and traversals with 50+ programs to practice.
- Graph : Terminology, implementation, traversals and other important programs like BFS, DFS, MST, etc
- Heap : Max, Min, Binary, Binomial, Heap sort.
- Advance DS : Hashing, Trie, Segment Tree, Suffix Tree

Beginner Algorithms



- Analysis of Algorithms : Time/Space Complexity, Asymptotic Analysis.
- Search : Linear, Binary.
- Sorting : Merge, Insertion, Selection, Counting, Radix.
- String Matching : Rabin Karp, etc

Advance Algorithms



- Divide and Conquer.
- Greedy Algorithms.
- Dynamic Programming.
- Branch and Bound.
- Backtracking.
- Randomized.

Learn the implementation and practice 4-5 questions each for all above mentioned algorithms.

Coding Question Lists



- DSA Cracker Sheet by Love Babbar ([link](#)).
- SDE Study Plan by Raj Vikramaditya ([link](#)).
- DSA Interview Questions on InterviewBit ([link](#)).
- Topic Wise Important questions on Leetcode ([link](#)).

Learning CP



- Start from beginner level on CodeChef. Practice at least 20-30 questions.
- Jump to next level, do 20-30 and so forth upto Hard.
- Check code of top submissions, compare your approach with them and see how you can further optimize.
- Focus on covering all use cases then rating.

Doing CP



- Participate in contests on CodeChef, monthly, annually, of different companies.
- Once you do 3-5 contests, join CodeForces, participate in contests there.
- Continue CodeChef contests on side, along with various coding competitions of top colleges/companies.
- You may refer to resources like A2oJ Ladders, CP Algorithms to study material related to CodeForces.

Where to study?



- For CP : Leetcode, GeeksForGeeks, CP Algorithms, A2oj Ladders, CodeChef, CodeForces.
- Only DSA : Hackerrank, GFG, Leetcode.
- YT Channels : MyCodeSchool, Abdul Bari, Neso Academy, Saurabh Shukla, Apni Kaksha.
- Books : Algorithms Made Easy by Narsimha Karumanchi

Tathastu DSA Series in 60 days @ Rs.500 (9456056603)



- 300+ programs covering every single topics of DSA on Leetcode using C++/JAVA/Python.
- Individual doubt support via Whatsapp groups.
- Daily LIVE doubt classes you can ask problems covered in the course.
- Course completion certificate with lifetime access to study material and code for each program in C++/Java/Python.

Register here :
<https://bit.ly/TathastuDSASeries>

Sample Problems



- Valid Palindrome ([link](#)).
- Print a 2D array in a spiral form ([link](#)).
- Next Greater Element ([link](#)).
- Merge 2 Sorted Linked Lists ([link](#)).
- Maximum Depth of Binary Tree ([link](#)).



Q & A

Let's hear what you guys have to say or

Email us at achintya@twowaits.com